



Rapport

– Project *Estate Mind* –

4DS5 – Group 5

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I. Business Understanding

Context

Real estate is a key sector in Tunisia, affecting households, investors and economic actors alike.

At the international level, this sector is undergoing a major transformation driven by PropTech platforms based on data and artificial intelligence.

In Tunisia, such approaches remain limited, which creates a real opportunity for local innovation.

It is within this context that our project takes shape: **Estate Mind**.

Problem Statement & Target Users

The **Tunisian real-estate market** is experiencing **strong growth**, with annual returns exceeding **9%**. However, despite this apparent dynamism, the process of buying, selling, or investing in property remains **complex** and **risky** for many stakeholders.

For **buyers**, the main challenges lie in navigating a large number of heterogeneous listings, assessing their credibility, avoiding overpriced properties, and identifying potential hidden defects or legal issues before committing to a purchase.

Investors face a different but equally critical set of difficulties. They must monitor emerging neighborhoods, analyze market trends and seasonal effects, anticipate future value appreciation, and minimize exposure to risky projects or unprofitable investments. The lack of consolidated and reliable market intelligence makes these strategic decisions particularly uncertain.

Sellers, on the other hand, often struggle to determine a fair and competitive price for their assets. Conflicting valuations, limited access to comparable transactions, and concerns about regulatory compliance can delay transactions and reduce market efficiency.

These difficulties are further amplified by several structural characteristics of the Tunisian market, including fragmented and heterogeneous data sources, strong regional disparities in pricing, the coexistence of transactions in dinars and foreign currencies, pronounced tourism-driven seasonality, and a complex regulatory framework related to land ownership and foreign investment.

In this context, the central question of the project can be formulated as follows:

How can we optimize real estate decisions in a fragmented and regulated market?

BO & DSO

BO (Business Objective)	DSO (Data Science Objective)
BO1 — Improve Market Reliability Increase transparency and quality of real-estate listings.	DSO1: Automate multi-source data ingestion.
	DSO2: Structure and clean listings using NLP/LLMs.
	DSO3: Track, version, and orchestrate data pipelines.
BO2 — Understand Territorial Dynamics Identify spatio-temporal trends and emerging areas.	DSO1: Model market activity with time-series methods.
	DSO2: Perform spatial aggregation and mapping of listings.
	DSO3: Detect emerging zones and trigger alerts.
BO3 — Estimate Prices with Confidence Deliver accurate and explainable property valuations.	DSO1: Predict prices using supervised learning models.
	DSO2: Integrate temporal and territorial drivers.
	DSO3: Explain predictions using XAI techniques.
	DSO4: Detect under- and over-valued properties.
	DSO5: Expose valuations and explanations through APIs and visual analytics.

BO4 — Support Investment Decisions Optimize choices through personalized analytics.	DSO1: Build user profiles from preferences and behavior.
	DSO2: Simulate profitability and “what-if” scenarios.
	DSO3: Recommend properties and strategies via scoring models.
BO5 — Secure Legal Compliance Reduce regulatory risk and automate conformity checks.	DSO1: Model legal rules in a regulatory knowledge graph.
	DSO2: Continuously update regulatory data.
	DSO3: Compute an automated legal-risk score.
	DSO4: Detect non-compliance and issue alerts.
	DSO5: Orchestrate specialized agents for continuous monitoring.
BO6 — Make the Platform Operational Enable access through AI assistants and automated reporting.	DSO1: Allow interaction via a conversational assistant.
	DSO2: Automatically generate reports and summaries.
	DSO3: Ensure explainability and traceability of outputs.

II. Data Acquisition & Understanding

Matching Data and Objectives

Business Objective	Key Data Needed	Purpose	Sources
BO1 — Market Reliability	IDs, timestamps	Ensure data quality & detect anomalies	Tayara, Mubawab...
BO2 — Territorial Dynamics	Location, coordinates, time, POIs	Analyze spatial & temporal trends	open data
BO3 — Price Estimation	Price, surface, rooms, location	Build predictive models	Tayara, Mubawab...
BO4 — Investment Support	Price, rental data, user preferences, economic data	Simulate ROI & recommend properties	open data
BO5 — Legal Compliance	Regulations, zoning rules, descriptions	Detect risks & non-compliance	banks , simulated data
BO6 — Platform Operation	All processed data, models, logs	Enable AI assistant & reporting	our solution

Data Sources Selection and Justification

To analyze the Tunisian real estate market, several major online real estate platforms were selected as primary data sources.

Selected Platforms

Source	Type	Reason for Selection	Limitations
Tayara	Classified platform	Large number of listings and wide geographic coverage	Noisy, duplicates
Mubawab	Real estate portal	Structured listings with detailed information	Medium Volume

Remax	Real estate agency	Professional listings and reliable information	Low Volume
Tecnoca sa	Real estate agency	Agency-based listings with market insights	Limited coverage
Century 21	International agency	High quality professional listings	Limited coverage
TPS Immobilier	Local agency	Additional coverage of local properties	Heterogeneous

Justification

These platforms were selected because they:

- represent a significant portion of the Tunisian real estate market
- contain relevant information such as price, location, and surface
- cover multiple regions and property types

Using multiple sources improves **data diversity, coverage, and reliability**.

Data Collection Implementation

Data collection is implemented using **web scraping and data ingestion techniques**.

A dedicated **Collector Agent (Agent Crawler)** is responsible for retrieving information from the selected platforms.

The crawler performs the following tasks:

- navigating platform pages
- extracting structured and semi-structured data
- collecting property attributes
- storing extracted information in the central database

Example of collected variables

Variable	Description
price	property price
surface	property surface area
property_type	apartment, villa, land, etc.
location	city and governorate

description	textual description of the property
publication_date	listing publication date
source	original platform

The collected data are stored in a centralized database used as the **knowledge base** of the system.

Global Architecture and Data Pipeline

The architecture of the Estate Mind system is based on a **multi-agent architecture**. The architecture was simplified to include **four main agents**.

System Architecture

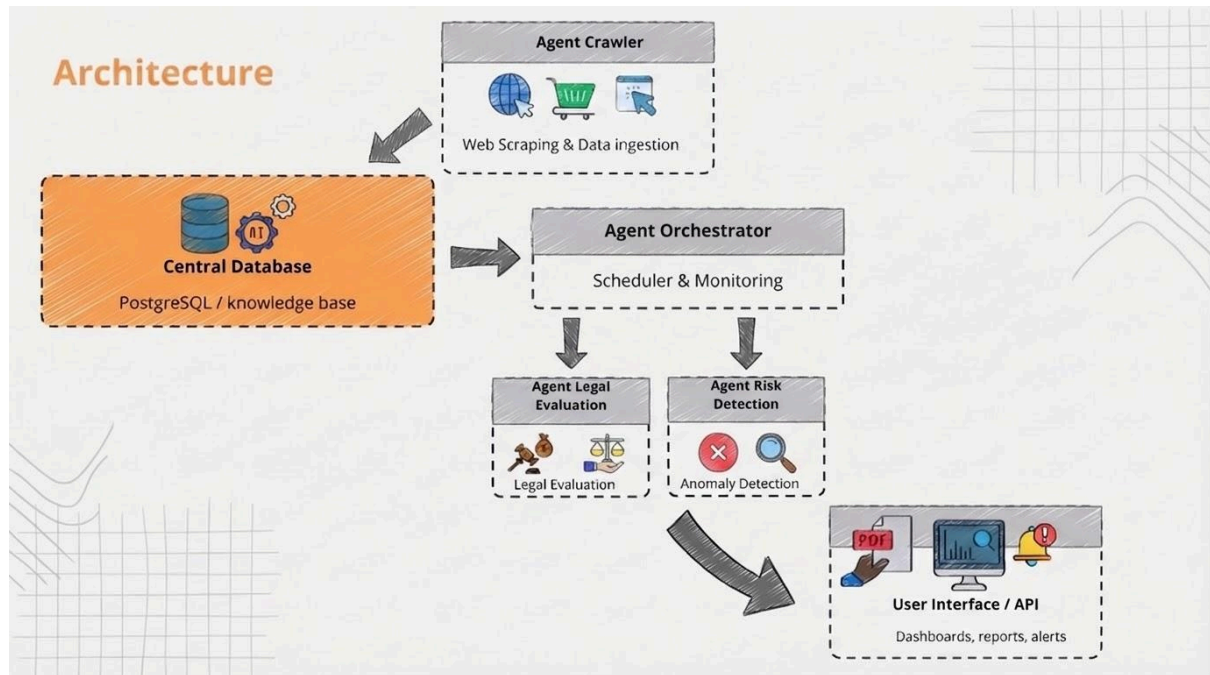
Agent	Role
Collector Agent	Collects data from real estate platforms via web scraping
Orchestrator Agent	Schedules tasks and monitors the data pipeline
Risk Detection Agent	Identifies anomalies and suspicious listings
Legal Agent	Evaluates regulatory and legal aspects of properties

The collected data are stored in a **central PostgreSQL database**. Each Agent is independent and has its own **knowledge base**.

The outputs generated by the system are exposed through a **User Interface and API**, enabling:

- dashboards
- automated reports
- alerts and notifications

This architecture ensures a modular and scalable system.



Data Understanding

An initial data understanding step was conducted to analyze the structure and characteristics of the collected datasets.

This analysis includes:

- identifying available variables
- analyzing data types
- detecting missing values
- identifying potential inconsistencies

Example of Main Variables

Variable	Type	Description
price	numeric	listing price
surface	numeric	property surface area
property_type	categorical	apartment, house, land
governorate	categorical	location at regional level
description	text	textual listing description
source	categorical	platform origin

Observed Data Issues

During the exploration phase, several common issues were identified:

- duplicate listings across multiple platforms
- missing values in surface or location
- inconsistent formatting of locations
- extreme price values or unrealistic listings

These observations guided the cleaning and validation steps.

Data Quality Validation and Initial Processing

The collected raw data were transformed into **quality data** through several cleaning and validation operations.

Cleaning Operations

The following transformations were applied:

- removal or treatment of missing values
- detection and removal of duplicate listings
- correction of inconsistent values
- standardization of formats (dates, units, text encoding)

Data Quality Indicators

Data quality was evaluated using the following criteria.

Quality Criterion	Description
Timeliness	We ensured data is up to date by tracking scraping dates
Comprehensiveness	We analyzed the coverage of all tunisian regions and property types
Relevance	We selected only features useful for our project objectives
Validity	We ensured values are plausible (price > 0, surface > 0)
Consistency	We harmonized the formats across sources
Accessibility	We structured storage enabling reproducibility

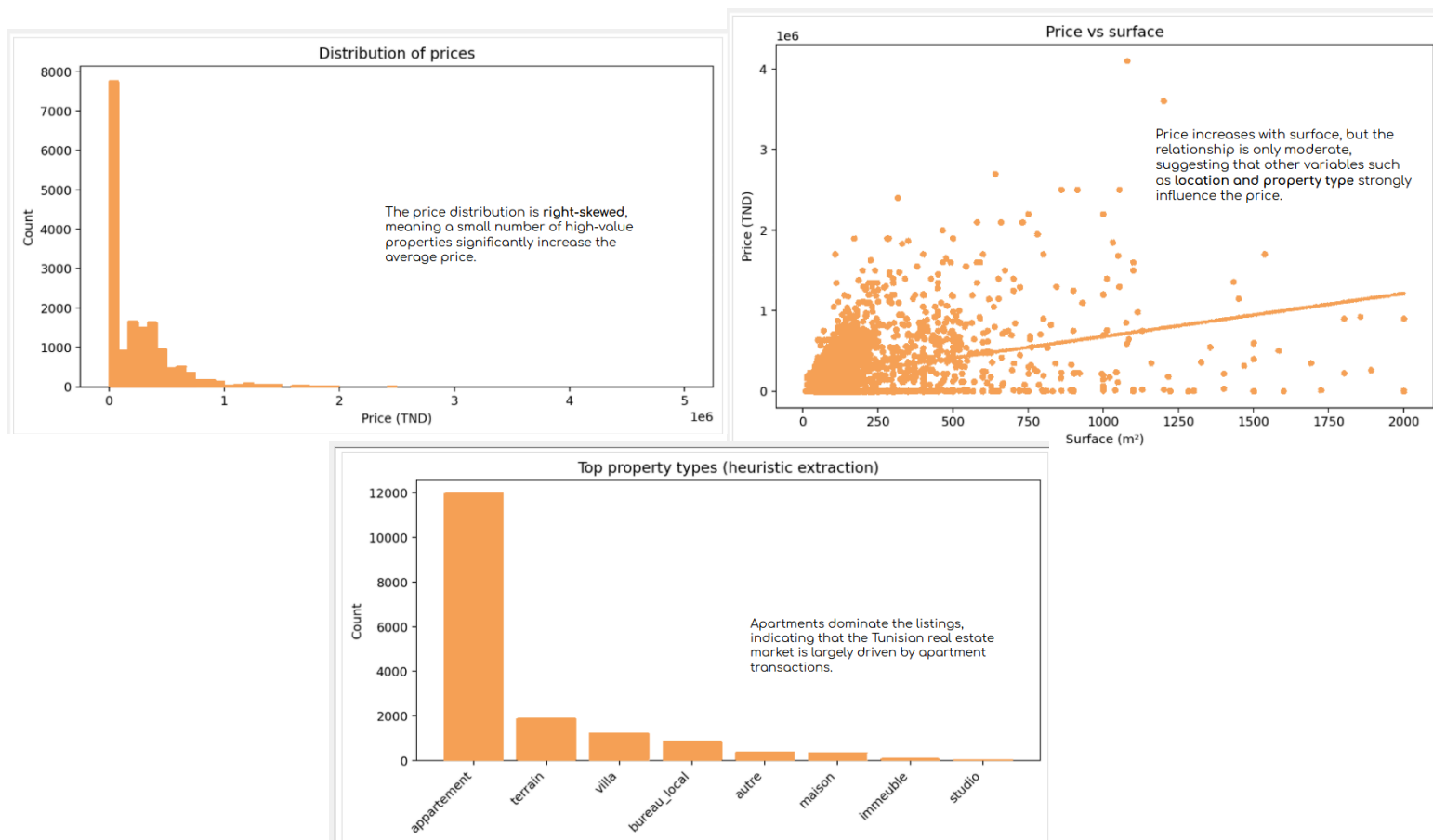
These checks ensure that the dataset is reliable and suitable for further analysis.

Exploratory Data Analysis (EDA)

An exploratory data analysis was conducted to better understand the characteristics of the dataset.

The EDA includes visualizations such as:

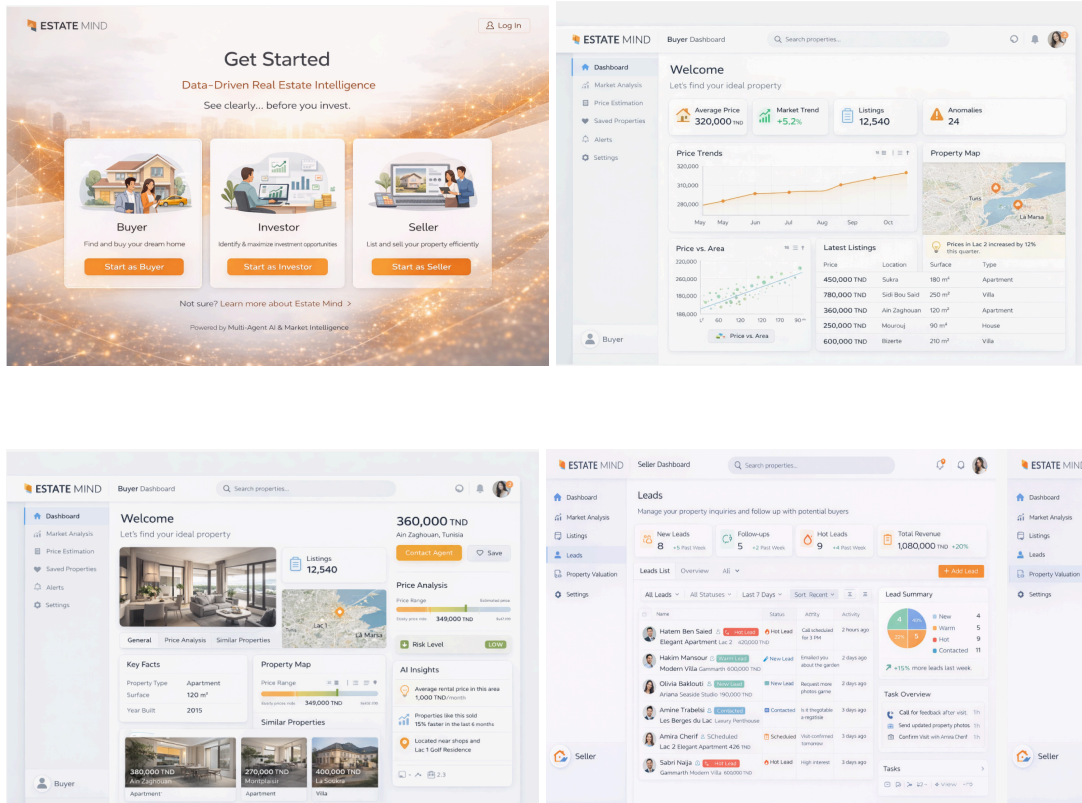
- price distribution
- relationship between price and surface
- price comparison across cities and governorates
- distribution of property types
- missing values analysis
- detection of outliers



These visualizations help identify patterns in the real estate market and highlight potential anomalies.

The complete EDA results are available in the **EDA notebook** and **HTML report**.

Mockups



Conclusion Phase 2

This phase allowed the project to establish a structured approach to data acquisition and understanding.

The main achievements of this phase include:

- identification and justification of relevant data sources
- implementation of an automated data collection process
- design of the system architecture and pipeline
- exploration and understanding of the collected data
- initial data cleaning and quality validation

These results provide a reliable dataset and a robust architecture for the next phases of the project, particularly data modeling, anomaly detection, and decision support for real estate investments.

III. Modeling

IV. Deployment